

SOLARSAFE - MECHANISM

Aditya and Donovan

OVERVIEW

- Team Management Plan
- Research
- Gaps in Solutions
- Project Objectives
- Changes in Scope
- Setup
- Experiments and Results
- Issues in Experiments and Actions to Take
- Personal and Technical Development
- Updated Timeline
- Plan for Completion

TEAM MANAGEMENT PLAN

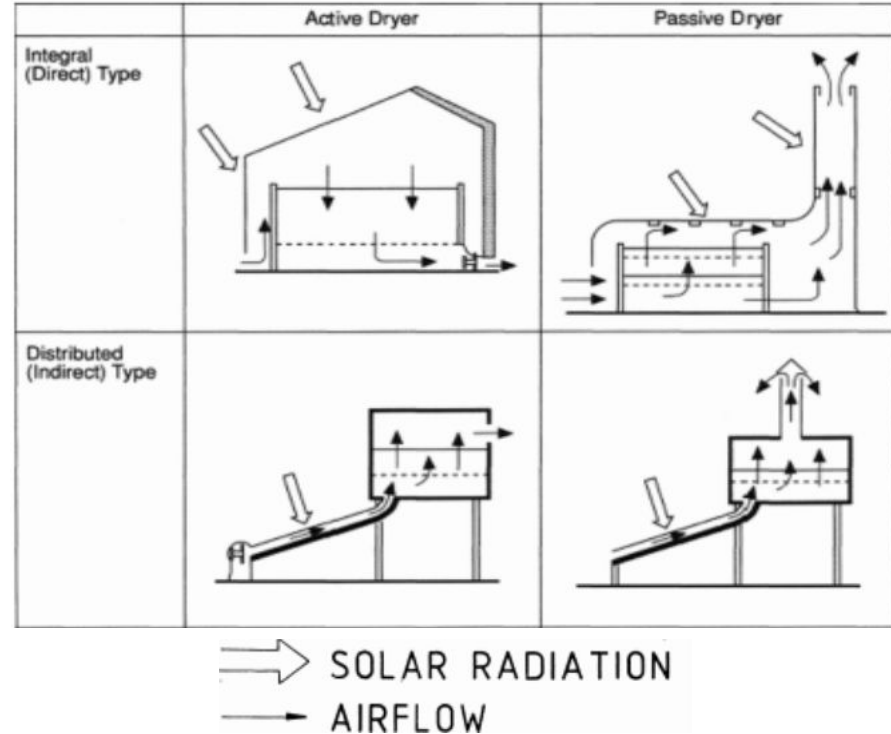
- Split into 3 sub teams
 - Better separation of concerns
 - More rapid development and ideation cycles
- Maintain mutual communication with each other
 - Minimise work overlap
 - Share resources and information

TEAM MANAGEMENT PLAN

- Workload allocation
 - Both of us were involved in all aspects
- Aditya
 - More experimentation-focused
- Donovan
 - More research-focused

RESEARCH

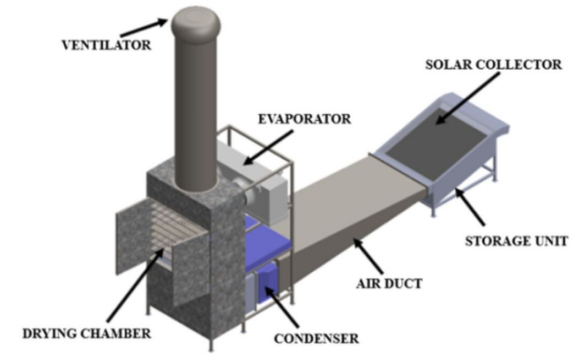
- Solar drying
 - Direct vs. Indirect
 - Active vs. Passive
- Implications:
 - Maize quality (discolouration, nutrient degradation, surface cracking)
 - Power consumption



(Ekechukwu & Norton, 1997)

RESEARCH

- Thermal Energy Storage (TES)
 - Phase-changing materials (paraffin wax, refrigerant)
 - Solids (sand, rocks)
 - Liquids (water)



(Kalage et al., 2026)



(Rulazi et al., 2023)

RESEARCH

- Burning fuels as a heat source
 - “Sustainable”
 - Still releases greenhouse gases

A) EasyDry M500 (495 kg, KES 85k)



B) Greenhouse dryer (180 kg, KES 33k)



C) Multipurpose dryer (90kg, KES 30k)



D) Solar POD dryer (90kg, KES 15k)



E) Dehytray (7 kg, KES 6k)



F) Hybrid Energy dryer (5 kg, KES 17k)



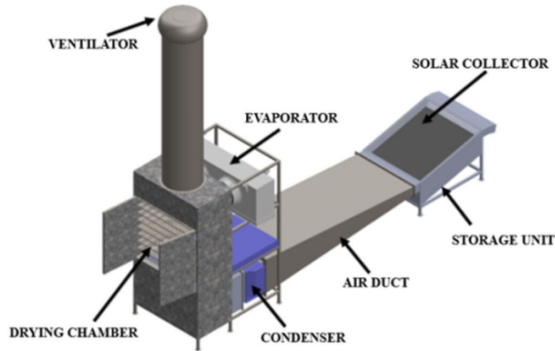
G) Traditional open-air drying (180 kg, KES 5k)



(De Groote et al., 2023)

GAPS IN SOLUTIONS

- Overengineered
- Relatively small capacity given cost
 - Need surface area to capture more solar energy
- Some use fuels



(Kalage et al., 2026)

G) Traditional open-air drying (180 kg, KES 5k)



(De Groote et al., 2023)

PROJECT OBJECTIVES

- Make solar dryer concept which is buildable by farmers
 - No specialised skill required
 - Easily sourceable materials
 - Easily repairable
- Can dry large amounts of maize with low cost
 - High drying to cost ratio
- No fuels burned

CHANGES IN SCOPE

- Reduced number of components
 - Limited experimentation rounds, one for day and one for night
- Focus on accessible and cheap materials
 - Farmers can build and scale up
- Explore usage of water bladder for TES
 - Direct conversion to heat for night-time

SETUP

- Black bin bag to line bottom and absorb sunlight
- Clear bin bag for greenhouse effect
- Air outlet (specified length for repeatability)
- Fan
- Clip to fasten bag to fan (without blocking airflow and allowing reusability)
- Shelter to protect fan from rain
- Whole cobs (can sit on mesh)

EXPERIMENT 1

- Whole cobs in standard setup recreation
- 1PM to 7PM
- Hot, and low humidity and cloud cover

avg temperature_2m (°C)	avg relative_humidity_2m (%)	avg cloud_cover (%)
26.71428571	36.28571429	59.14285714

EXPERIMENT 1



EXPERIMENT 1 RESULTS

- Starting weight of 796g
- End weight of 754g
- Roughly 42g (5.28wt%) lost by water evaporation
- 1 or 2 corn kernels lost, but a single kernel <1g so should be negligible

EXPERIMENT 2

- Drying in an elevated, fully exposed position by placing on a mesh
- Parallel control experiment for comparability
- Measured masses of cobettes individually
- Much cooler, more humid and more cloud cover than previous day

avg temperature_2m (°C)	avg relative_humidity_2m (%)	avg cloud_cover (%)
21.51428571	46.14285714	95.42857143

EXPERIMENT 2



EXPERIMENT 2 RESULTS

- The corn on the mesh showed just under 40% more drying, despite being placed behind the control cobettes

31/05	1100	1730	mesh 1	116	104	10.34%
			mesh 2	80	72	10.00%
			mesh 3	92	84	8.70%
			mesh 4	102	96	5.88%
			mesh avg.			8.73%
			control 1	114	108	5.26%
			control 2	114	106	7.02%
			control 3	98	92	6.12%
			control 4	118	110	6.78%
			control avg.			6.30%

ISSUES IN EXPERIMENTS AND ACTIONS TO TAKE

- Movement of shade with sun's position
→ Place in centre of garden and move preemptively
- Battery depletion after 6 hours (claimed 9 hours)
→ Additional battery packs?
- Black bin bag partially blocking air flow
→ Fix in place
- Curvature from inflation causing black bin bag to curl and cast shade on the corn
→ Reduce length

PERSONAL AND TECHNICAL DEVELOPMENT

Aditya

- Experimental design with lack of controlled factors
- Design considerations and building in an agile way
- Pivoting plans when plans no longer feasible
- Being resourceful and communicative to get things done

Donovan

- Clearer articulation of thoughts and ideas
- Narrowing of scope so that it is commensurate with time and manpower available
- Seeing the bigger picture; future plans for project
- Familiarisation with Dyson Centre and equipment

UPDATED TIMELINE

Week 3 (remaining):

- Trials of water bladder as TES system for night drying

Week 4:

- Build and test chimney
- Analysis of results and conclusions

PLAN FOR COMPLETION

- Obtained experimental evidence for efficacy of modifications
- Given starting point for integration into SOLARSAFE system
 - Mesh table concept
 - TES
- Presented ideas for scalability

THANK YOU!

Hope you enjoyed!

Check out our setup and samples + Questions